

Lecture 01

Introduction to Environmental Economics

Ivan Rudik

AEM 4510 / AEM 5510 / ECON 3865

Roadmap

- Course set up
- What is environmental economics?
- What are the goals for this class?
- Microeconomics recap

Course set up

Course logistics

Class: Tuesday and Thursday, 2:55–4:10 p.m., Mallott 228

Professor: Ivan Rudik (ir229)

Professor office hours: Tuesday, 10:00–11:00 a.m., Warren 462

TA office hours:

- Chrisma D'Souza (ccd84): Friday, 1:00–2:00 p.m., Warren 330
- Jacob Holifield (jh2799): Monday, 3:00–4:00 p.m., Warren 330

Office hours begin in Week 2

Readings and prerequisites

Required text: *Markets and the Environment* by Keohane and Olmstead

Other readings: papers linked on the course website or available through the Cornell library or Google Scholar

Optional econometrics reference: *Causal Inference: The Mixtape* by Cunningham is available free online

Prerequisites: MATH 1110 and either AEM 2600 or ECON 3030

Grading

4 problem sets (12%): 3% each

4 in-class prelims (68%): 17% each

Optional final (20%):

- If you take all four prelims, you may skip the final
- If you skip it, your final-exam grade equals your average prelim grade
- If you miss a prelim, the final is required and also replaces the missed prelim grade, so it counts for 37% of your course grade

Letter grades: A = 93–100 and A- = 90–92; the remaining cutoffs follow the syllabus through D- = 60–62, with F below 60

Problem sets

Due at 11:59 p.m.: Sep 10, Oct 8, Nov 5, and Dec 1

- One problem set is due before each prelim
- Work in teams of up to three; submit one PDF per team through Gradescope
- Include every group member's name
- You may submit as often as needed before the deadline
- No late work: solutions are posted immediately after the deadline
- Your final submission before the deadline is the version that will be graded

Prelims and final

In-class prelims: Sep 15, Oct 15, Nov 10, and Dec 3

- Prelims emphasize material since the prior prelim, but may use all earlier material
- The optional final is cumulative; its date is to be determined
- Bring only a pen or pencil: paper is provided, and calculators, notes, and electronics are not allowed
- Problem sets and the additional review questions on Canvas are the main preparation

Missed prelims

Your first missed prelim: no makeup, regardless of the reason

- Its 17% weight moves to the final, so the final becomes 37% of your course grade
- The final is no longer optional

Later missed prelims: a valid excuse is required for a makeup

Student Disability Services

This course participates in the SDS Alternative Testing Program

- Request your approved testing accommodation for this course as soon as possible
- Submit an Exam Request Form for each exam through the SDS student portal
- SDS will email the time and room no later than 48 hours before the exam
- Send exam-logistics questions to sds-testing@cornell.edu
- Contact me about access barriers or non-testing accommodations

Content

Weeks 1-7ish: the economics of environmental regulation

- More theoretical, graphical
- Taxes, cap and trade, regulating monopolies, tariffing other countries, etc

Weeks 7-14ish: using markets (real and financial) to learn about the environment

- Some theory, also real world analyses and data
- valuing recreation, housing markets, bond markets, green instruments

In class

No recording or sharing: do not record, photograph, or share classroom interactions or course materials without written permission

No electronics: laptops, tablets, phones, and other electronics are prohibited; take notes on paper

Exceptions are available for approved accommodations. Students permitted to record must still protect the privacy of the classroom.

Why no electronics?

The reason for no electronics is pedagogical: **to improve learning in the course.**

Laptop or tablet use in class (even just for notes) is associated with decreased learning and lower test scores in a variety of settings

I post slides after we finish each section so you can review the material

Sana, F., Weston, T., & Cepeda, N. (2013). *Computers & Education*, 62, 24-31.; Mueller, P. & Oppenheimer, D. (2014). *Psychological Science*, 25(6), 1159-1168.; Carter, S. P., Greenberg, K., & Walker, M. S. (2017). *Economics of Education Review*, 56, 118-132.

Last words

Canvas contains the full syllabus, assignment instructions, readings, and review materials

Attendance is not required, but it is highly recommended

Grade appeals must be submitted through Gradescope within 24 hours

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Any questions about the course?

What is environmental economics?

What's enviro econ?

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These ideas can be applied to the environment

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Environmental economics helps us understand things like:

The value of mitigating pollution

How agents will response to climate change policies

Whether investment tax credits for wind power are cost-effective

Air pollution is bad



How do people respond to info?



ADVISORY: [#ozone](#) is expected to be Unhealthy for All today in and around the foothills area of [@SequoiaKingsNPS](#) (Ash Mountain entrance)



Ozone Air Quality Guide

Advisory Level	Health Concern
Unhealthy (86 ppb – 105 ppb)	The following groups should avoid prolonged outdoor exertion: • People with lung disease, such as asthma • Children and older adults • People who are active outdoors Everyone else should limit prolonged outdoor exertion.

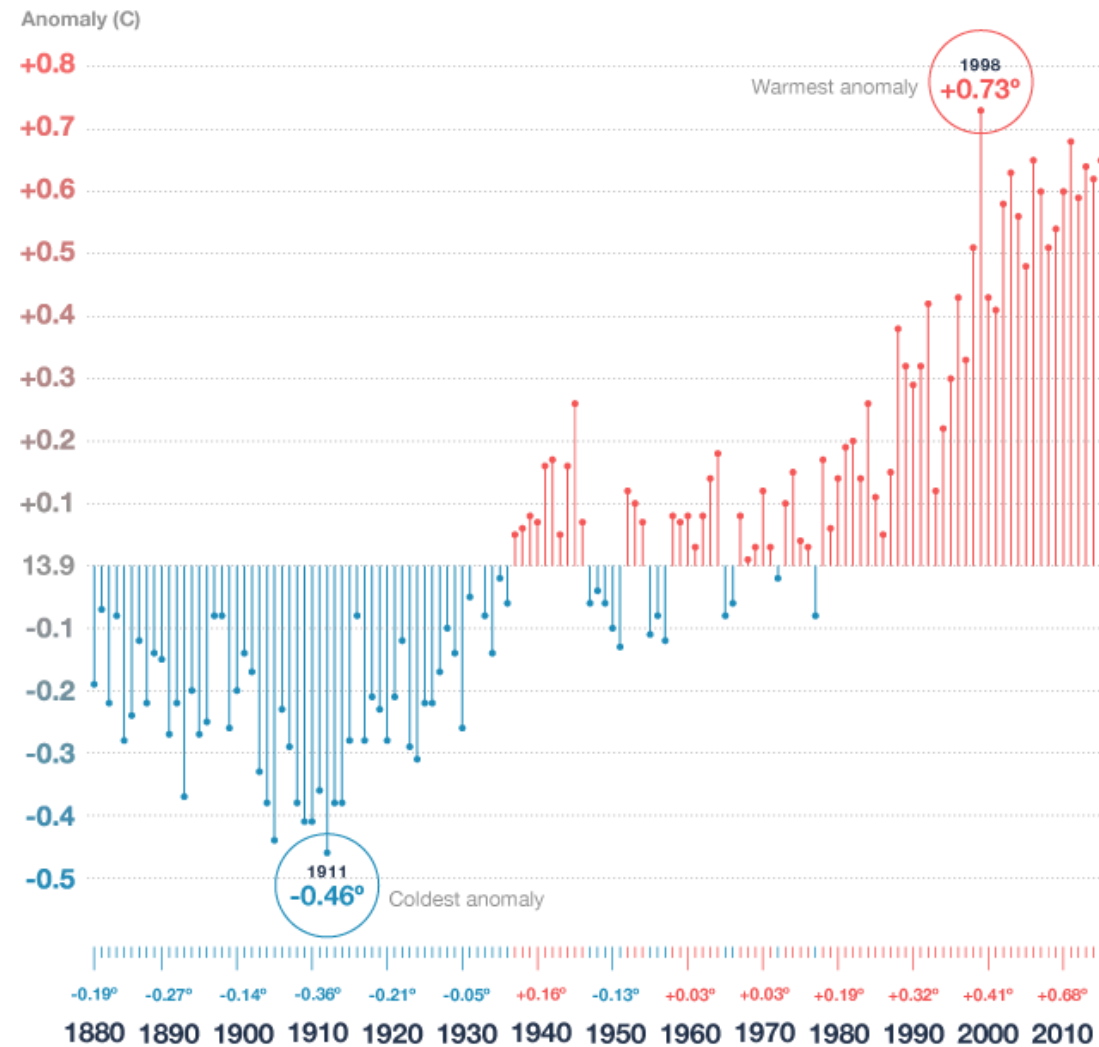
The ozone advisory level is based on an 8 hour average ozone concentration; ppb = parts per billion.

Places provide info to help people avoid air pollution

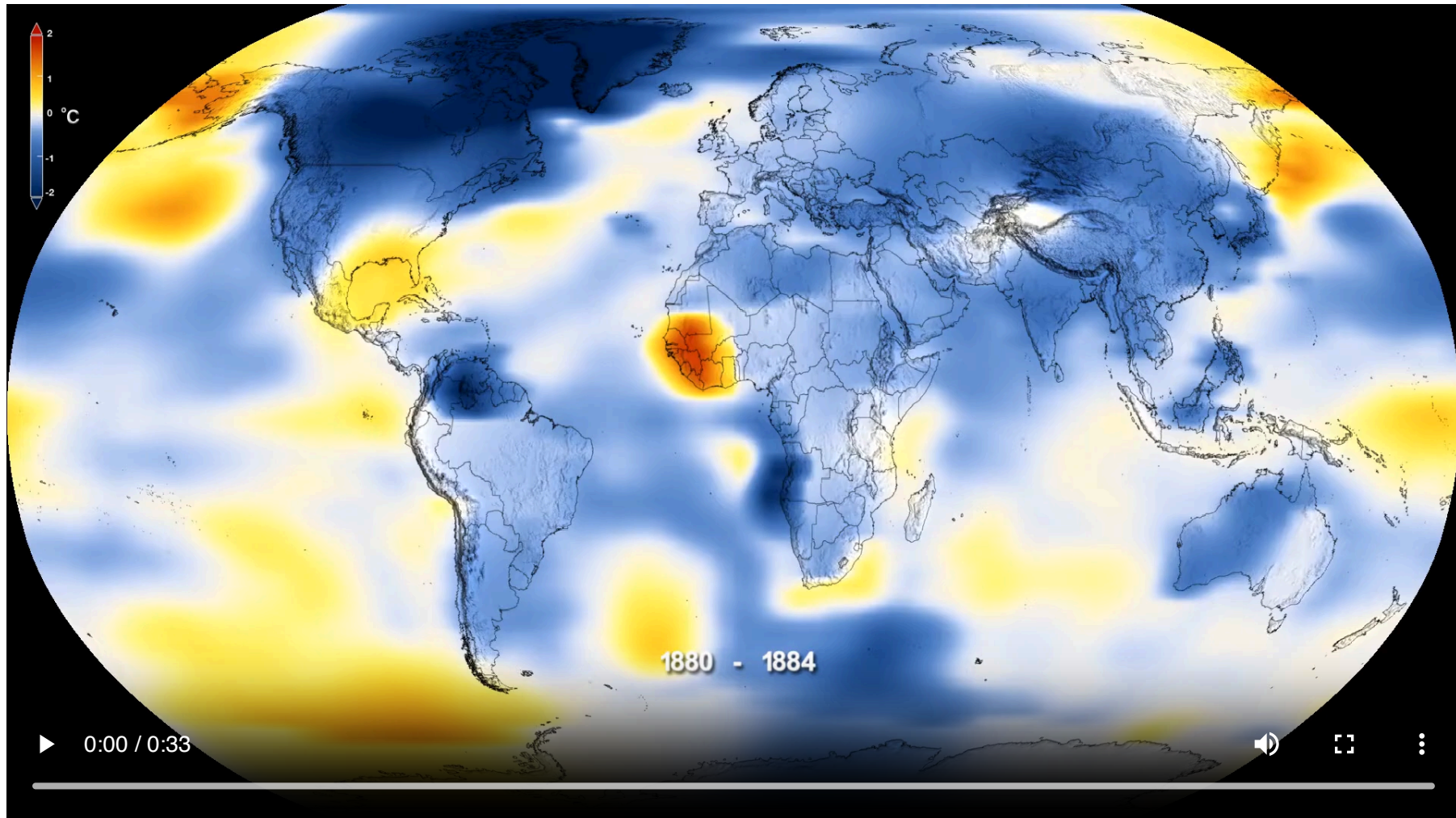
Does it work?

How well?

Climate change



Climate change



Climate change

Why do economists care about climate change?

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- Production
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- Fishing

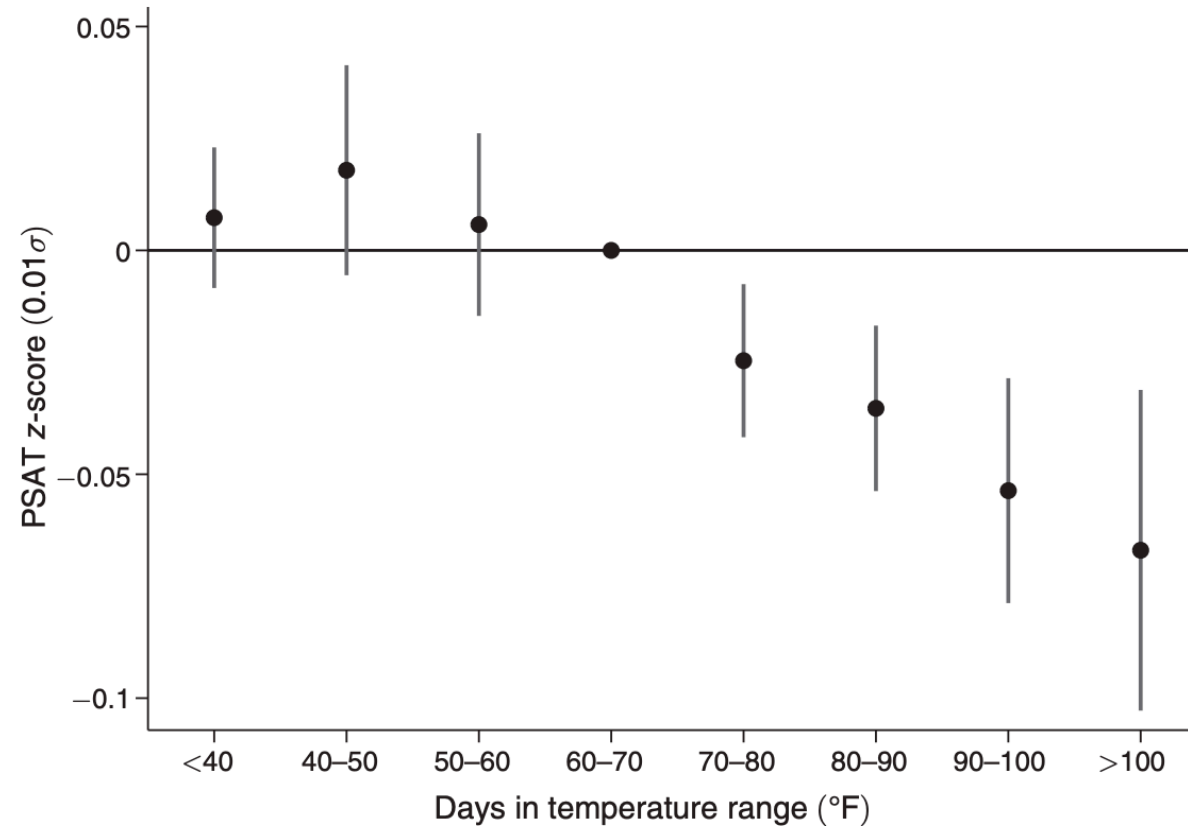
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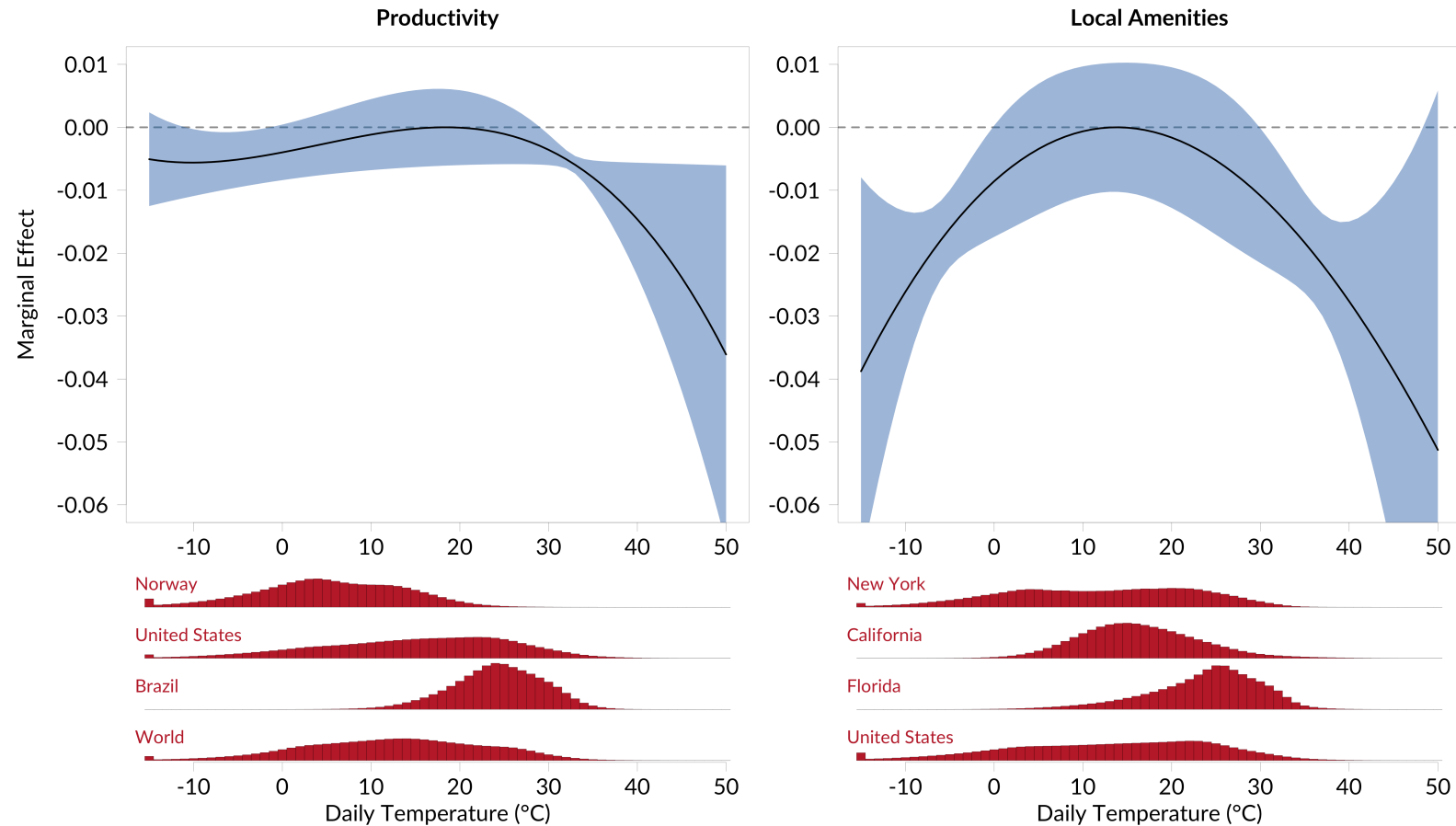
- Production
- Learning
- Leisure
- Fishing
- etc, etc

Climate change: heat hurts learning



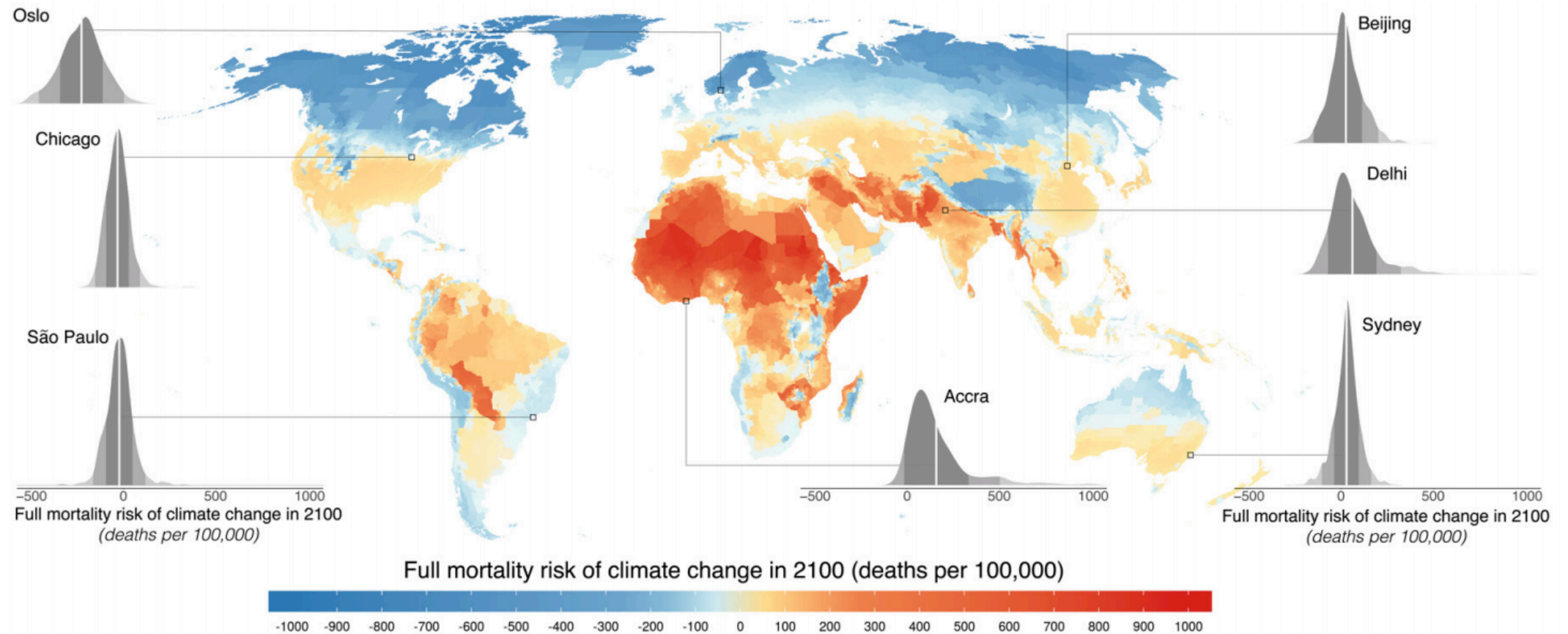
Park et al. (2020)

Climate change: heat hurts the economy



Rudik et al. (2021)

Climate change: extreme heat/cold increases mortality

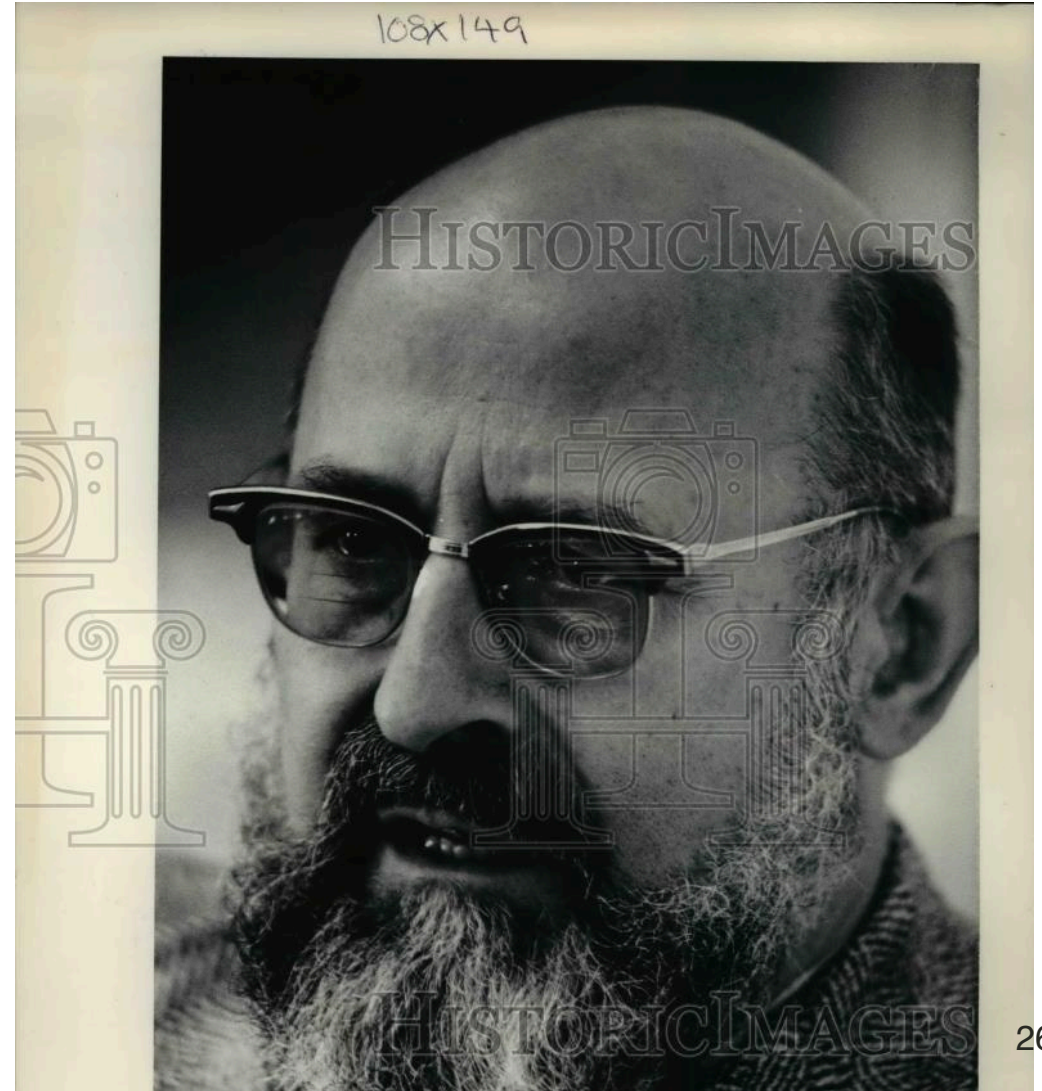


What is enviro econ?

Environmental economics is actually pretty new

Spurred by **John Krutilla** in the 1950s

His paper **Conservation Reconsidered** is the landmark paper in the field (sort of like *Wealth of Nations* and economics as a whole)



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- Industry jobs (energy, transportation, finance)
- Public sector and NGOs (EPA, DOE, RFF, Brookings, Federal Reserve)
- Graduate programs

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Learn both the theory and **applications** of environmental economics

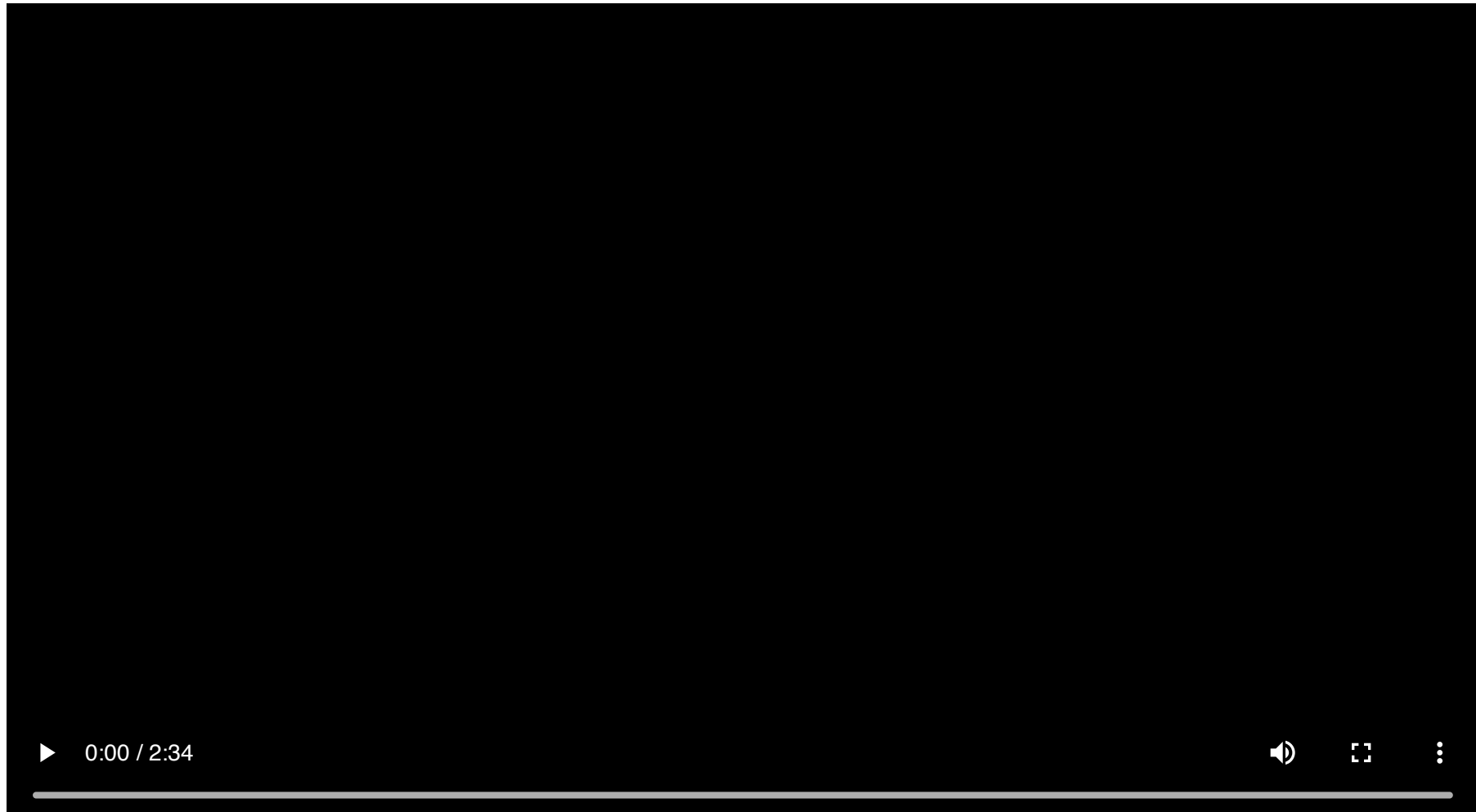
Microeconomics recap

Micro recap

Is Intro to Micro applicable everywhere?

Micro recap

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Creating markets to solve problems

How do we solve some problems in practice?

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Micro recap: terminology

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Supply curve:

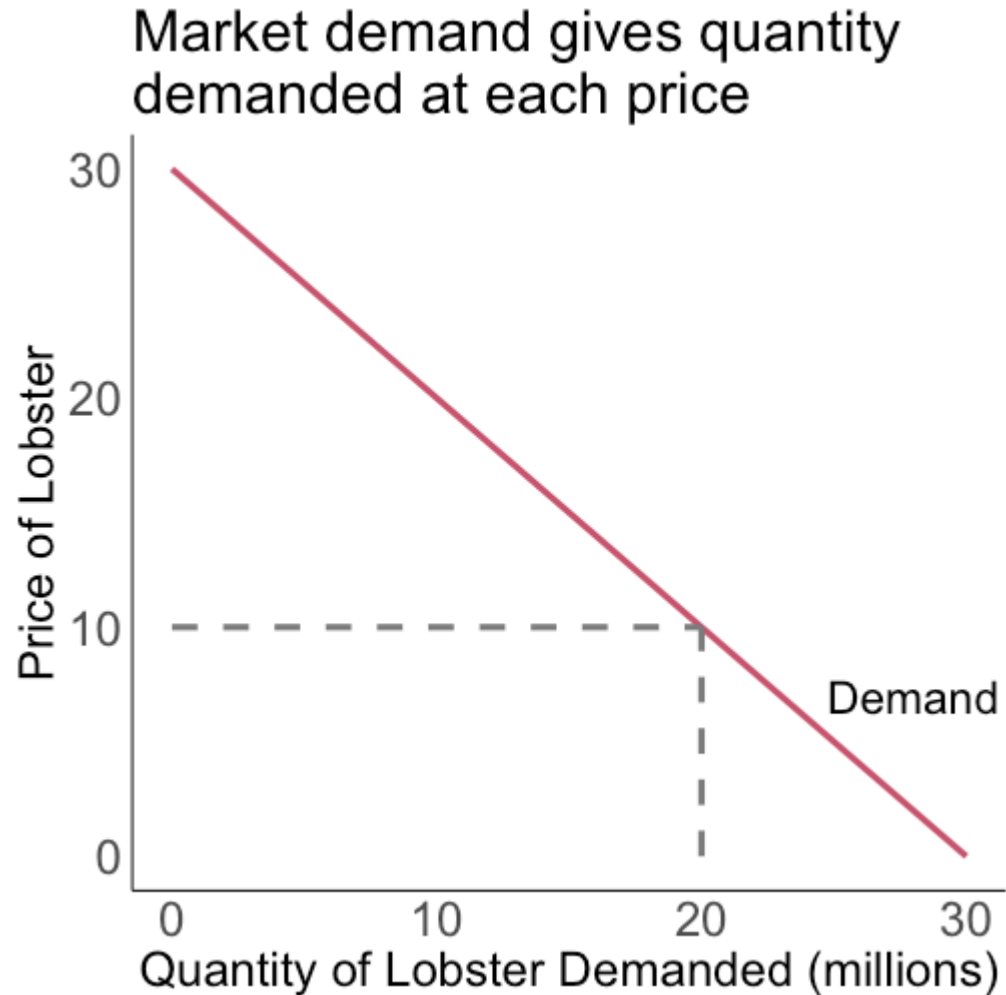
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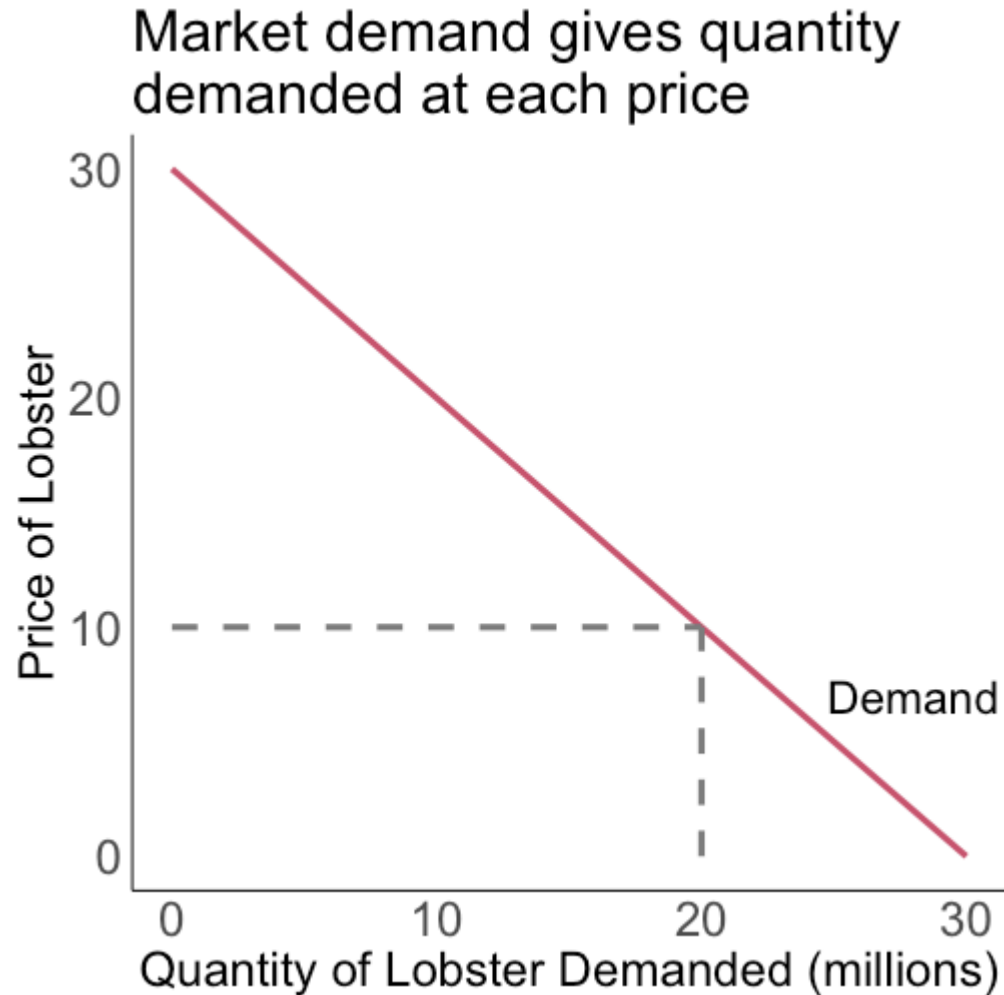
Supply curve: A schedule or graph showing the quantity of a good that sellers wish to sell at each price; it gives us the marginal willingness to accept or the marginal cost

Market demand



Market demand is aggregated from all individual demand curves

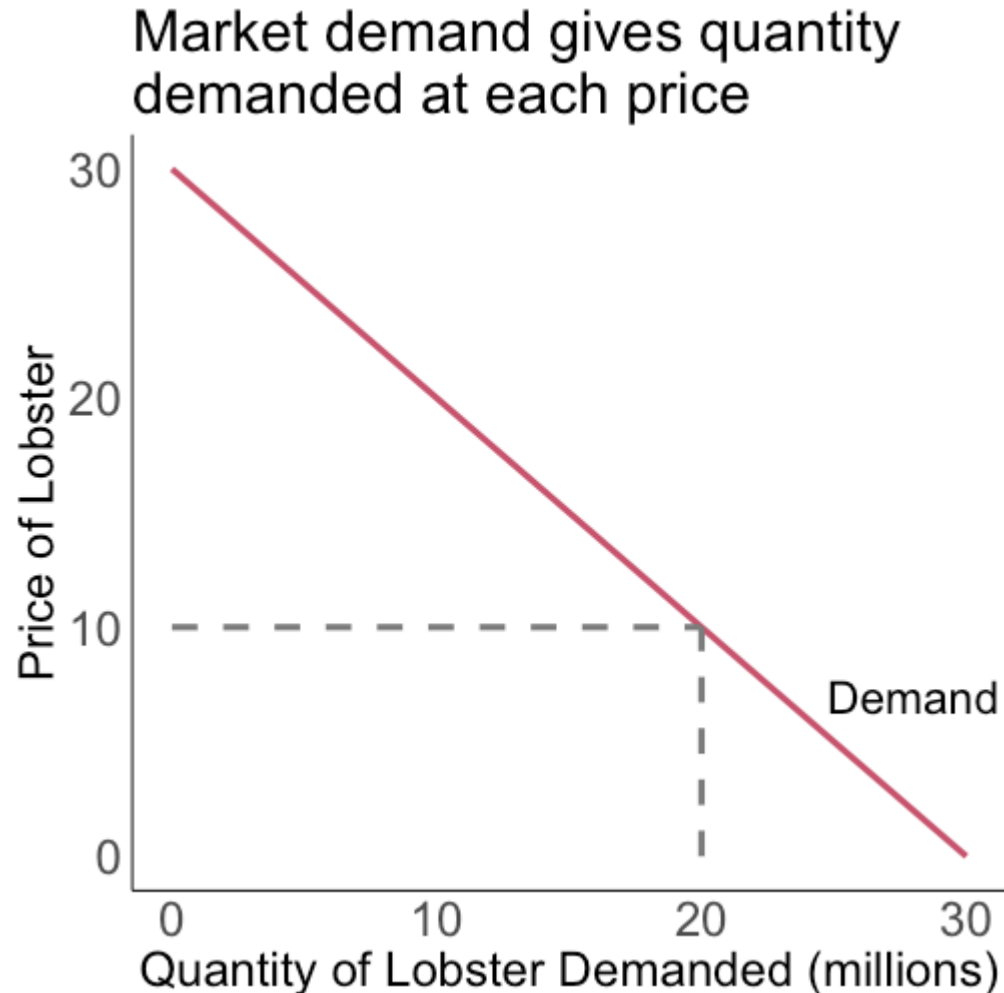
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Horizontal interpretation: if buyers face a price of \$10/lobster they will want to purchase 20 million

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Vertical interpretation: if buyers are buying 20 million lobsters, the marginal buyer is willing to pay at most \$10

Market demand slopes down

Why do demand curves slope down?

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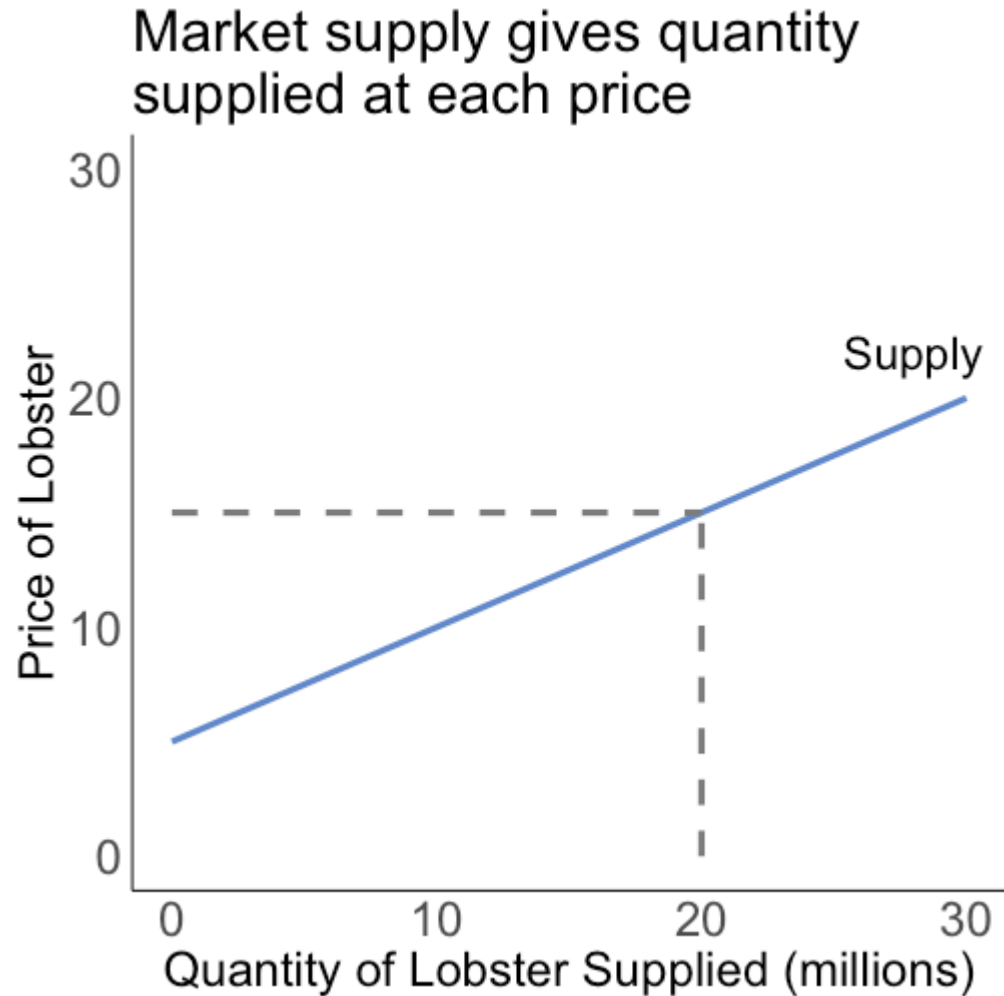
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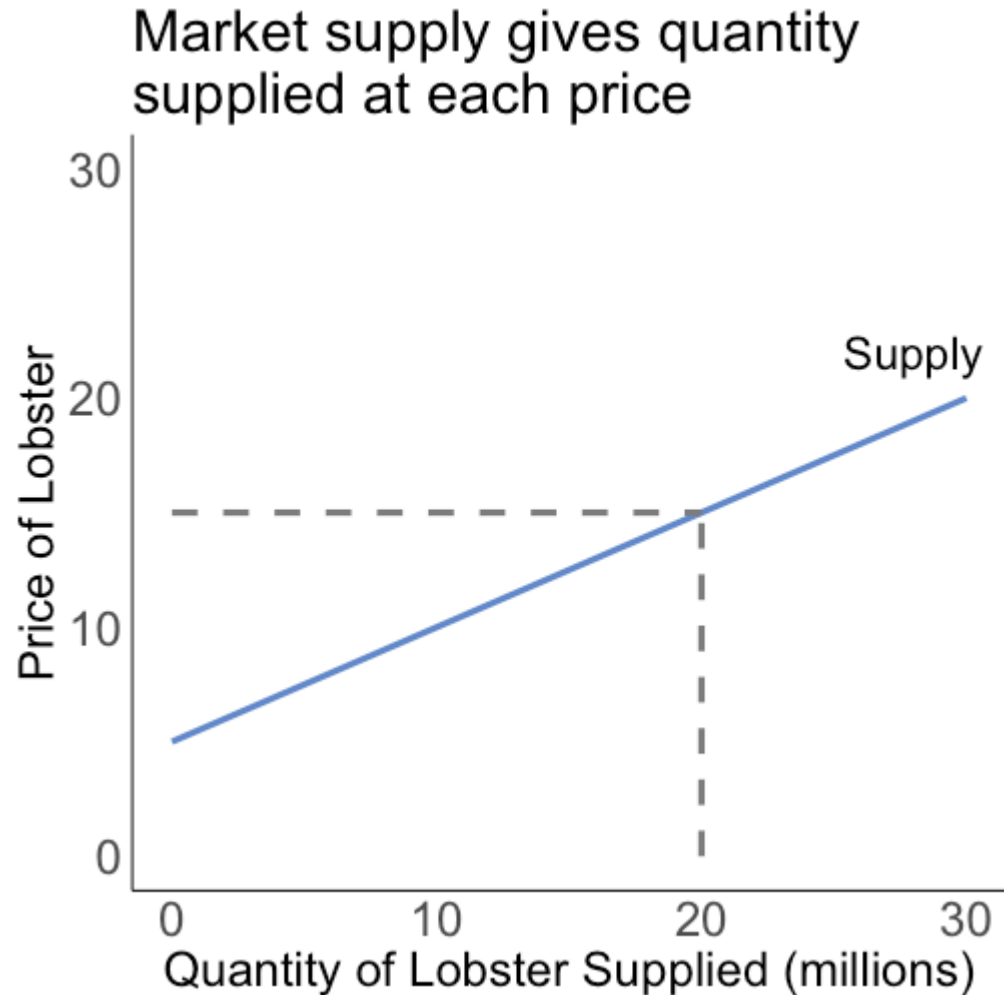
This is the **income effect**: if the price of pizza goes up, we have a lower real budget

Market supply



Market supply is aggregated from all individual supply/MC curves

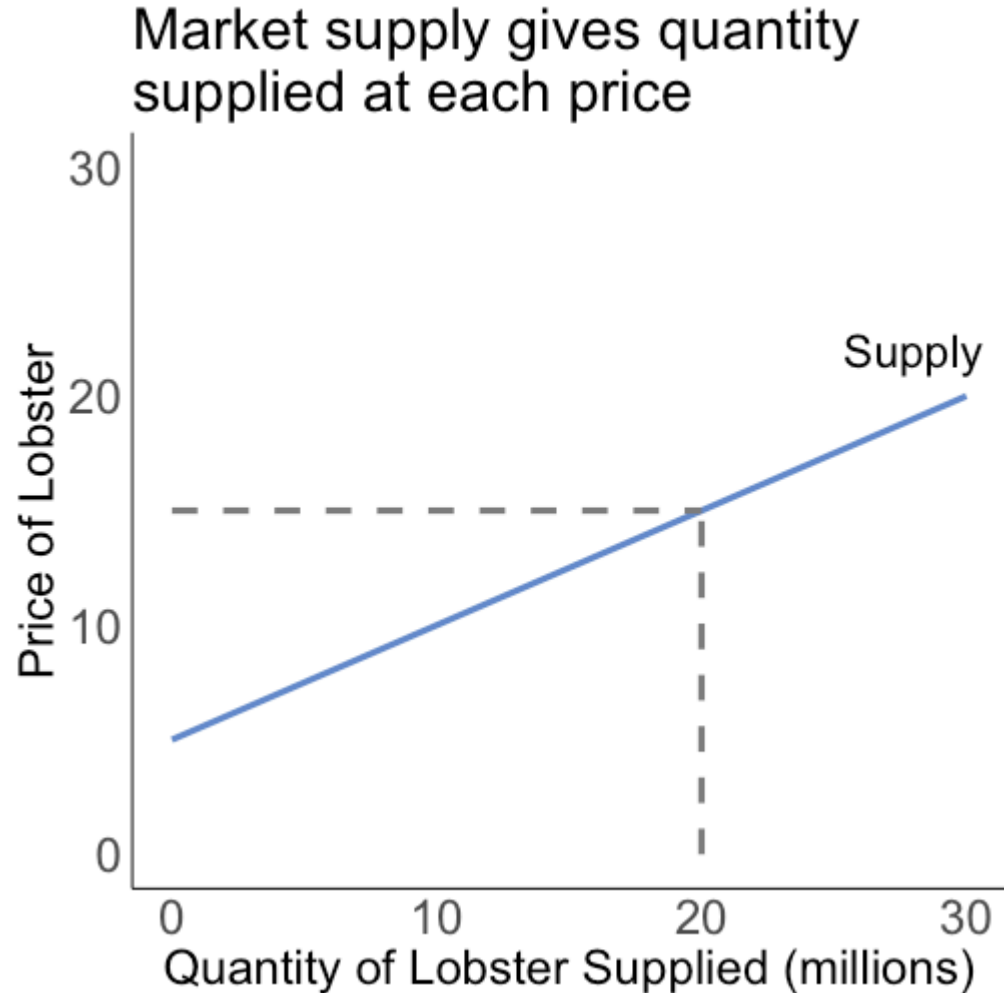
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Vertical interpretation: if sellers are selling 20 million lobsters, the marginal cost of the last lobster is \$15

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If we produce more of a good, we choose the lowest (opportunity) cost production processes first, higher cost production processes later

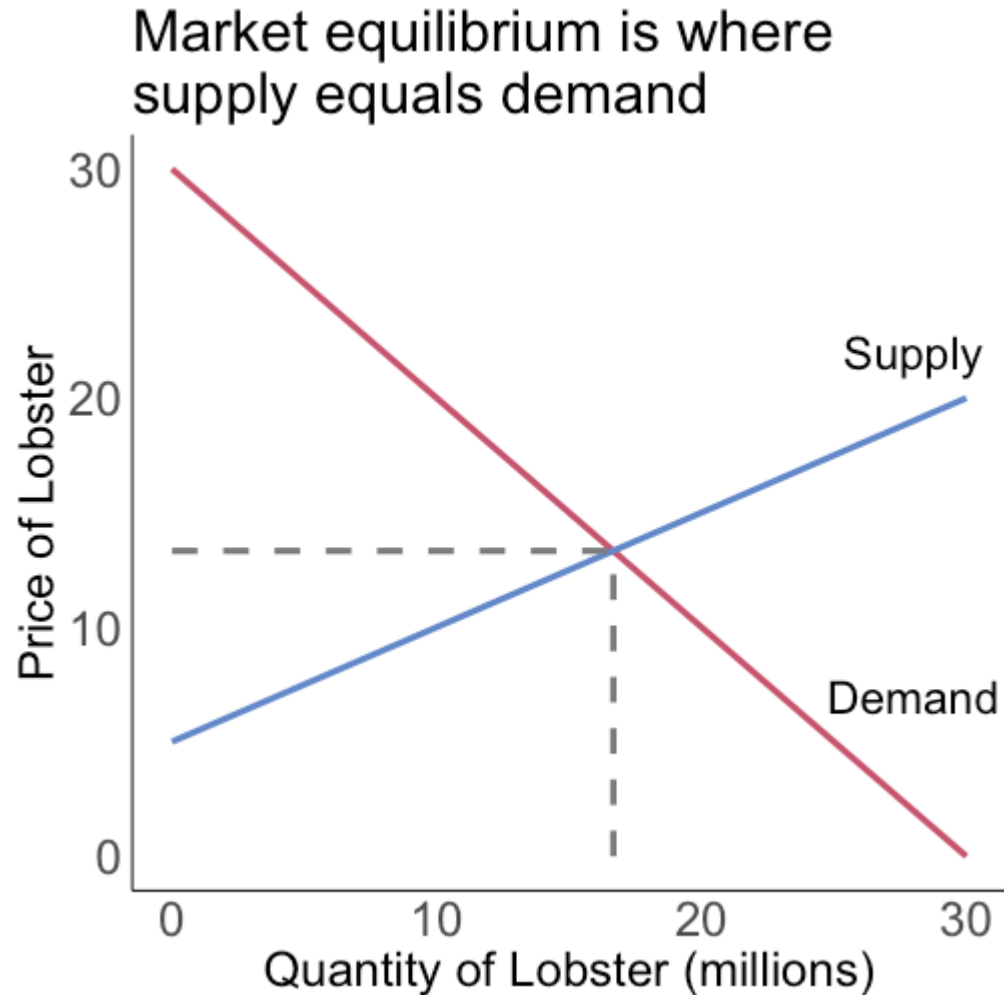
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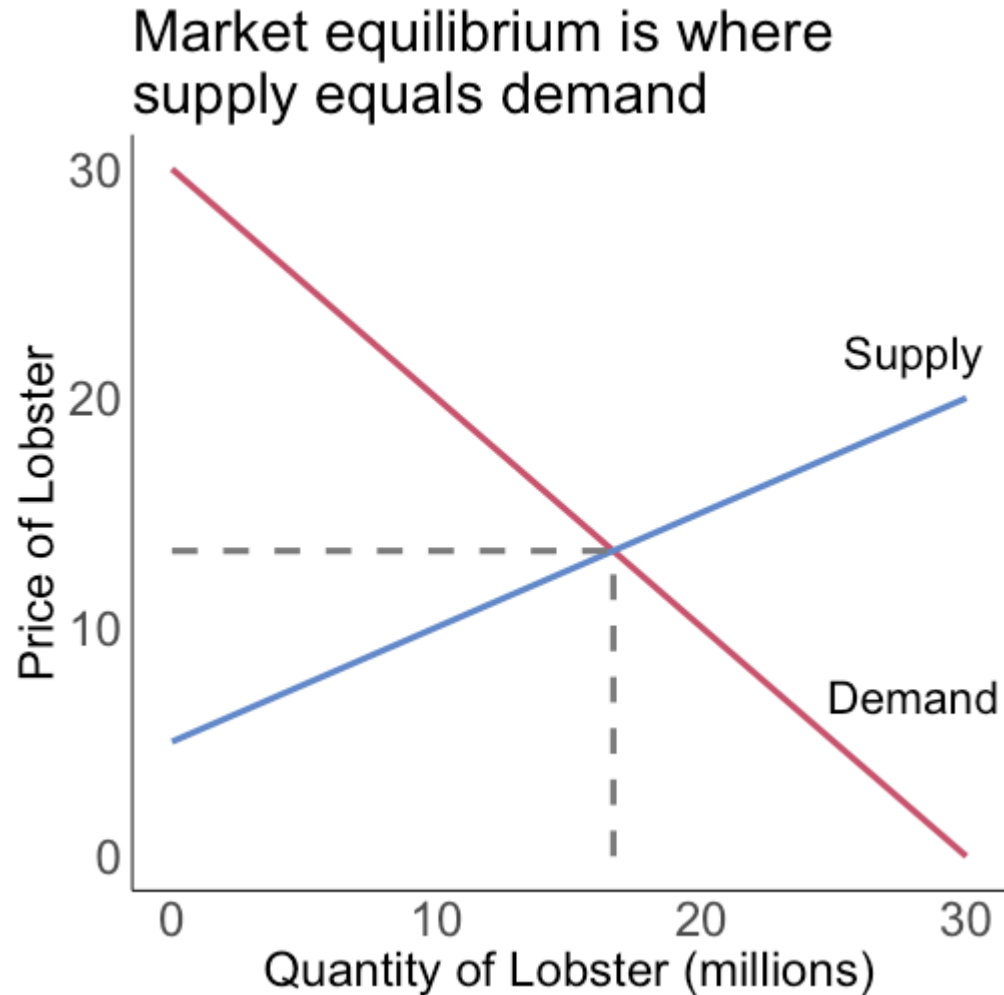
Marginal costs go up as production goes up → producers need higher prices in order to produce more goods

Market equilibrium



A market equilibrium is a price/quantity pair where the demand curve crosses the supply curve

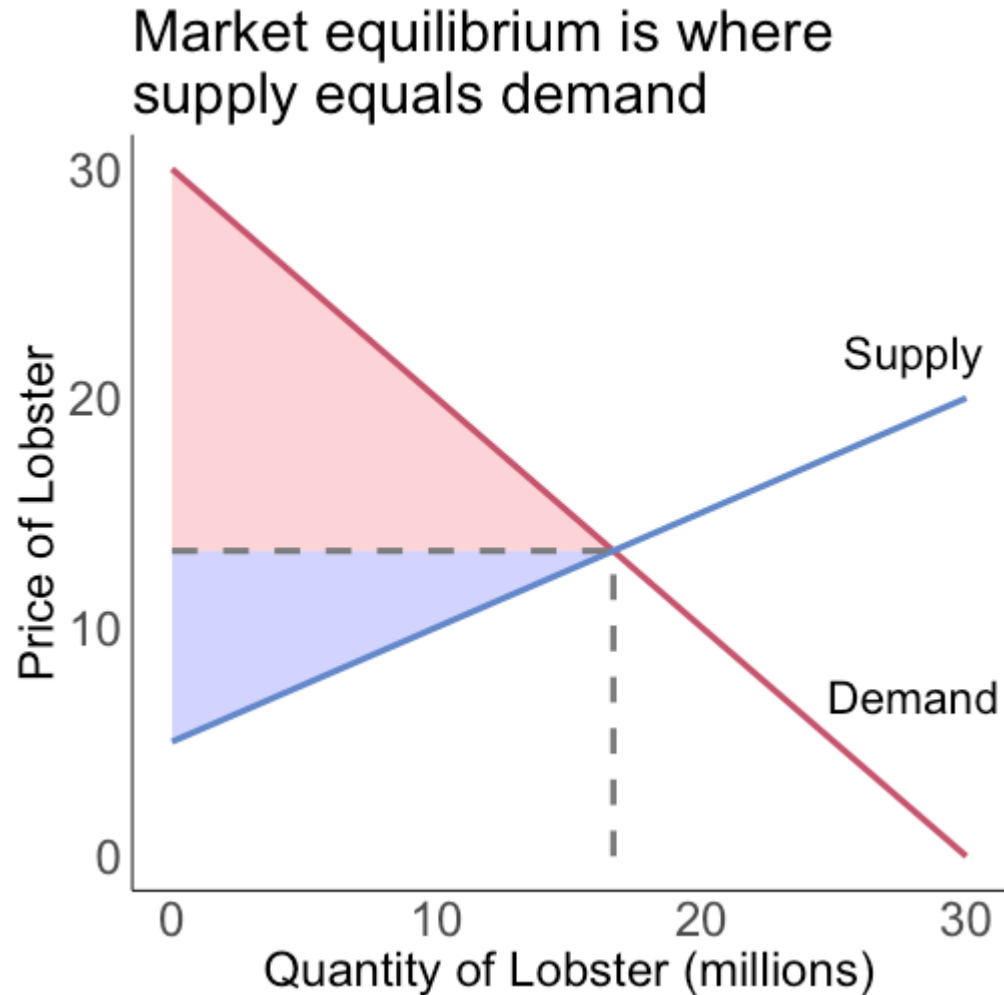
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This gives us the price where the quantity demanded exactly equals the quantity supplied: no shortages, no surpluses

Market equilibrium



Consumer surplus is the difference between willingness to pay (demand) and price

Producer surplus is the difference between price and marginal cost (supply)

Total surplus is the sum of CS and PS

Efficiency

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2. An allocation is **Pareto efficient** if there's no way to change things to make at least one person strictly better off, without making at least one person strictly worse off

If you can make one person better off without making anyone else worse off, it's called a **Pareto improvement**

Are markets efficient?

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Main takeaway: markets are often a nice way to allocate scarce resources

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3. Complete markets (minimal transactions costs)
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They drive a wedge between private and social marginal cost, or private and social marginal benefit

The key departures in environmental economics

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	Excludable	Non-Excludable
Rival	Private goods (food)	Common-pool resources (fish, timber)
Non-rival	Club goods (parks, netflix)	Public goods (air, national defense)

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Why?

Without the adequate information, buyers or sellers cannot make the choices in their best interest

Efficiency and equity

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- If Elon Musk held **all** the wealth in the world, that would be a Pareto efficient outcome even though it goes against basically all people's notions of equity
- Perfect price discrimination is also Pareto efficient: producers capture all the surplus but it is maximized

Ecological wealth

